Industrial Internship Report on

" Banking Information System"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship offered by **Upskill Campus** and **The IoT Academy** in collaboration with the industrial partner **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship was focused on developing a solution for a project/problem statement assigned by UCT. The project, along with its documentation, was completed within a period of six weeks. This internship provided an excellent opportunity to gain practical exposure to Core Java programming and software development practices. My project, **"Banking Information System,"** is a console-based application developed using **Core Java**. The system enables users to register bank accounts, log in using an account number, deposit and withdraw money, check account balances, update mobile numbers, and view transaction history. The project also incorporates object-oriented programming concepts, file handling, exception handling, and input validation to create a simple, secure, and user-friendly banking application. This internship gave me valuable exposure to real-world software development and helped me strengthen my technical skills in Core Java. It improved my understanding of object-oriented programming, problem-solving techniques, debugging, and project implementation. Overall, it was a highly enriching learning experience that enhanced both my technical knowledge and confidence in developing software applications.

TABLE OF CONTENTS

1 Preface

Industrial internships play a vital role in bridging the gap between academic learning and practical industry experience. They provide students with an opportunity to apply theoretical knowledge to real-world problems while improving their technical, analytical, and professional skills. This six-week internship conducted by **Upskill Campus** in collaboration with **The IoT Academy** and **UniConverge Technologies Pvt. Ltd. (UCT)** provided me with valuable exposure to software development using Core Java.

During this internship, I worked on the project **"Banking Information System,"** a console-based application developed using Core Java. The project was designed to simulate basic banking operations such as user registration, account login, money deposit, money withdrawal, balance enquiry, transaction history, and mobile number update. The application also incorporates object-oriented programming concepts, file handling, exception handling, and input validation to ensure efficient and reliable operation.

The internship was planned in a structured manner, beginning with Core Java concepts and gradually progressing toward project implementation. Weekly learning modules, quizzes, and progress reports helped me strengthen my understanding of Java programming and software development practices. These activities provided continuous learning and encouraged me to improve my coding and problem-solving abilities.

Throughout this internship, I enhanced my knowledge of object-oriented programming, collections, file handling, exception handling, debugging, and software design. I also developed confidence in building a complete Java application independently while following good programming practices.

I would like to express my sincere gratitude to **Upskill Campus**, **The IoT Academy**, **UniConverge Technologies Pvt. Ltd.**, and my faculty members for providing me with this valuable learning opportunity and continuous guidance throughout the internship.

Finally, I would like to encourage my juniors and fellow students to actively participate in internships and practical projects. Such experiences not only improve technical knowledge but also build confidence, problem-solving skills, and industry readiness, which are essential for a successful career in software development.

2 Introduction

An industrial internship is an essential part of a student's academic journey, providing practical exposure to real-world technologies and software development practices. It helps students apply theoretical concepts, improve problem-solving skills, and gain experience in developing industry-oriented applications.

As part of my six-week internship conducted by **Upskill Campus** in collaboration with **The IoT Academy** and **UniConverge Technologies Pvt. Ltd. (UCT)**, I worked on a **Banking Information System** using Core Java. The objective of this project was to develop a console-based application that simulates basic banking operations in a simple, secure, and user-friendly manner.

The application enables users to create bank accounts, log in using an account number, deposit and withdraw money, check account balances, update mobile numbers, and view transaction history. The project also demonstrates the use of Object-Oriented Programming (OOP), file handling, exception handling, collections, and input validation.

This internship enhanced my programming knowledge, strengthened my understanding of Core Java concepts, and provided practical experience in designing, implementing, testing, and debugging software applications. It also improved my confidence in developing complete Java-based projects independently.

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

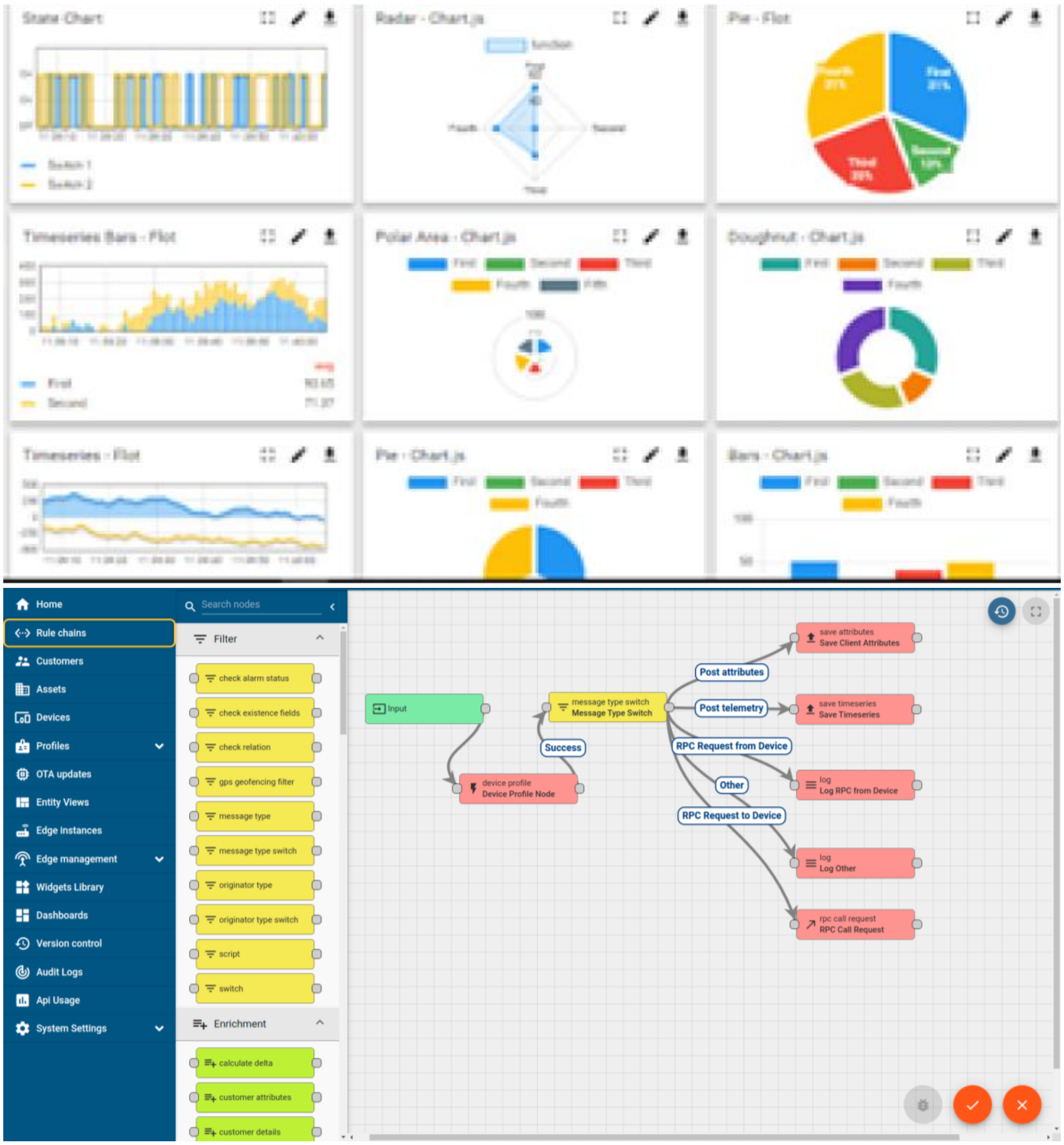
i. UCT IoT Platform ( )

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine


State Chart
Radar - Chart.js
Pie - Flot
Timeseries Bars - Flot
Polar Area - Chart.js
Doughnut - Chart.js
Timeseries - Flot
Pie - Chart.js
Bars - Chart.js
Home
Rule chains
Customers
Assets
Devices
Profiles
OTA updates
Entity Views
Edge instances
Edge management
Widgets Library
Dashboards
Version control
Audit Logs
Api Usage
System Settings
Search nodes
Filter
check alarm status
check existence fields
check relation
gps geofencing filter
message type
message type switch
originator type
originator type switch
script
switch
Enrichment
calculate delta
customer attributes
customer details
Input
message type switch
Message Type Switch
device profile
Device Profile Node
Success
Post attributes
Post telemetry
RPC Request from Device
Other
RPC Request to Device
save attributes
Save Client Attributes
save timeseries
Save Timeseries
log
Log RPC from Device
log
Log Other
rpc call request
RPC Call Request

ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

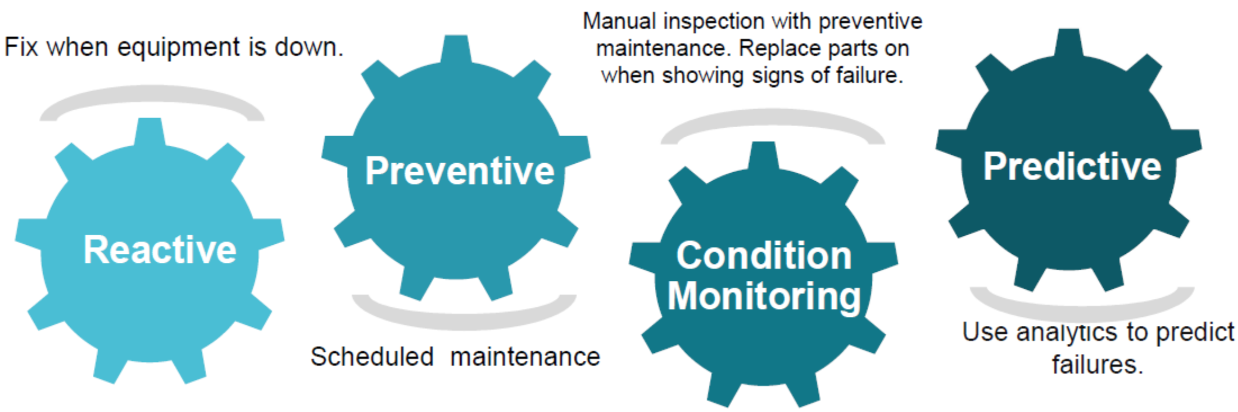
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

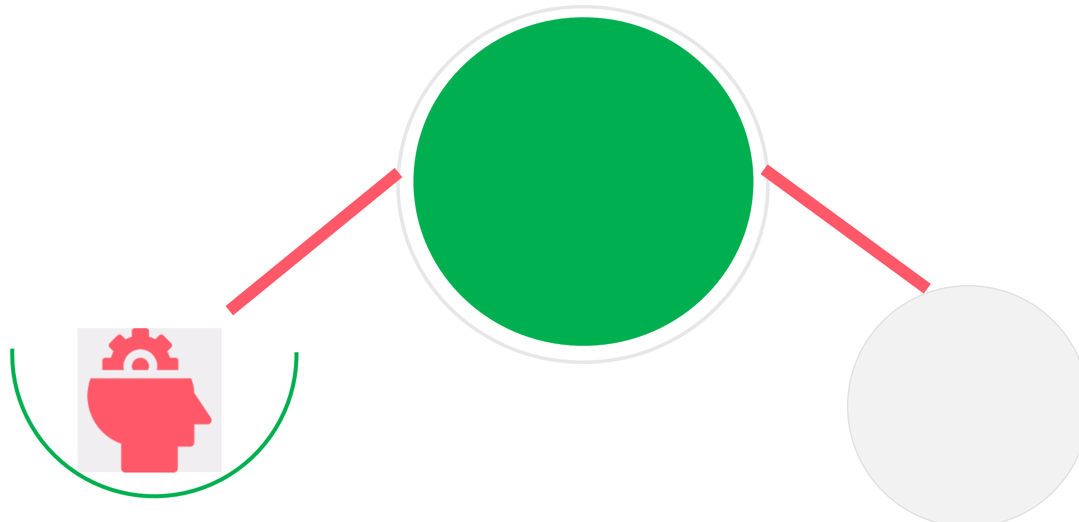
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

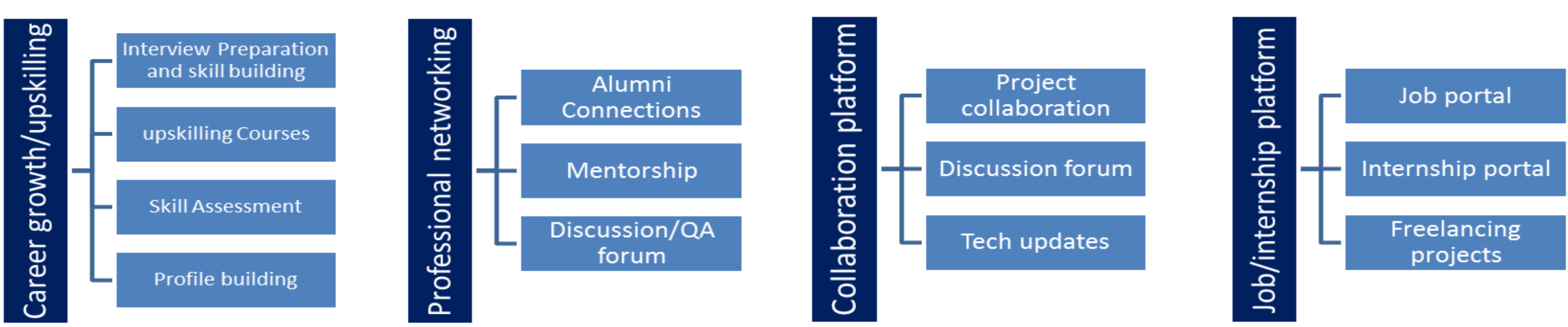
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 Objectives of this Internship program

The main objectives of this internship were:

- To gain practical experience in Core Java programming and software development.
- To understand and apply Object-Oriented Programming (OOP) concepts in real-world applications.
- To develop a complete Banking Information System using Core Java.
- To improve problem-solving, debugging, and coding skills through project implementation.
- To learn file handling, exception handling, collections, and input validation in Java.
- To gain experience in using Git and GitHub for project version control.
- To improve documentation and project presentation skills through report preparation.

2.4 Reference

1. Core Java learning materials provided by Upskill Campus.
2. The IoT Academy course resources.
3. Oracle Java Documentation – https://docs.oracle.com/javase/
4. Java Tutorials by W3Schools – https://www.w3schools.com/java/
5. GitHub Documentation – https://docs.github.com/

2.5 Glossary

Acronyms	Terms
OOP	Object-Oriented Programming
JDK	Java Development Kit
JVM	Java Virtual Machine
IDE	Integrated Development Environment
API	Application Programming Interface
VCS	Version Control System
GUI	Graphical User Interface
FH	File Handling

3 Problem Statement

Traditional banking operations often require customers to visit a bank branch or maintain manual records for basic transactions such as account creation, deposits, withdrawals, and balance enquiries. Managing these operations manually can be time-consuming, error-prone, and inefficient.

The objective of this project is to develop a **Banking Information System** using **Core Java** that automates these basic banking operations through a console-based application. The system allows users to create a bank account, log in securely using an account number, deposit and withdraw money, check account balances, update mobile numbers, and view transaction history.

The application also implements **Object-Oriented Programming (OOP)** concepts, **file handling**, **exception handling**, and **input validation** to ensure reliable data management and provide a simple, user-friendly experience. This project demonstrates the practical application of Core Java concepts while improving efficiency, accuracy, and usability in performing basic banking operations.

4 Existing and Proposed solution

4.1 Existing System

The traditional banking system relies on manual processes for maintaining customer records and performing transactions. This approach can be time-consuming, prone to human errors, and inefficient in managing account information. It also makes tracking transactions and updating customer details more difficult.

4.2 Proposed Solution

The proposed **Banking Information System** is a console-based application developed using **Core Java** that automates basic banking operations. The system allows users to register accounts, log in using an account number, deposit and withdraw money, check account balances, update mobile numbers, and view transaction history. It also uses file handling to store account information and exception handling to improve reliability.

4.2.1 Comparison of Existing and Proposed System

Existing System	Proposed System
Manual record maintenance	Digital record management using Core Java
Time-consuming operations	Fast and automated banking operations
Higher chance of human errors	Improved accuracy through automation
Difficult to track transactions	Transaction history is maintained
Manual customer record updates	Easy mobile number update feature
No proper data validation	Input validation and exception handling

4.3 Code submission (Github link)

https://github.com/tejomahima777/upskillcampus

4.4 Report submission (Github link) : To be updated after uploading the final report PDF to GitHub.

5 Proposed Design/ Model.

The proposed Banking Information System is a console-based application developed using Core Java. It follows the Object-Oriented Programming (OOP) approach by separating different functionalities into individual classes. The system provides a simple and user-friendly interface that allows users to perform banking operations efficiently.

The application consists of multiple modules, including account registration, login, deposit, withdrawal, balance enquiry, transaction history, and mobile number update. It also uses file handling to store account details and exception handling to prevent runtime errors. The modular design makes the application easy to understand, maintain, and enhance in the future.

5.1 High Level Diagram (if applicable)

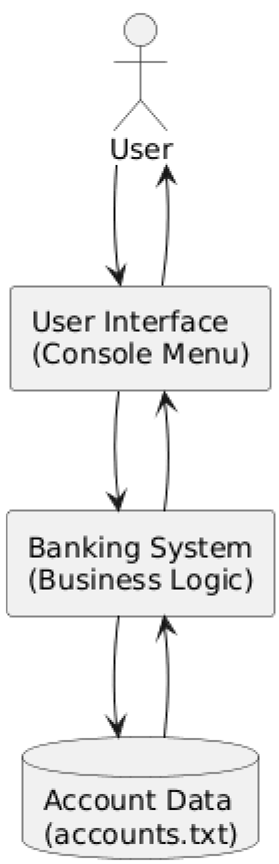


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

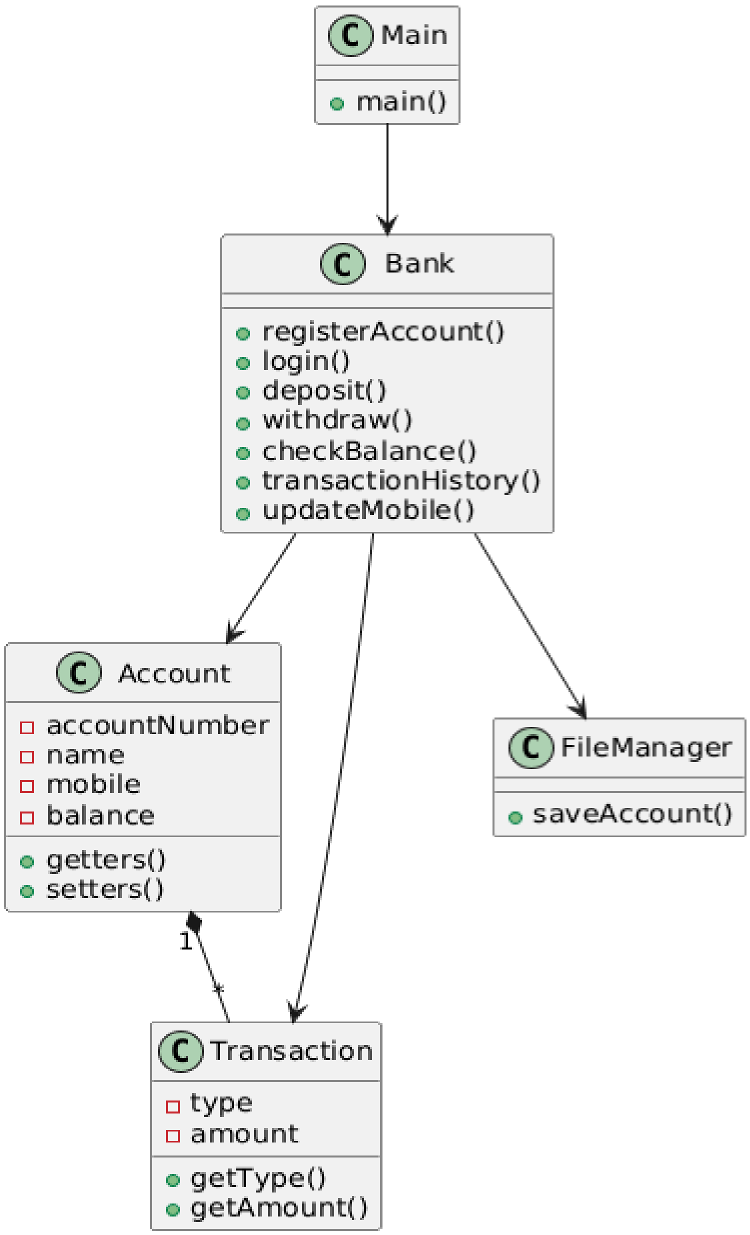
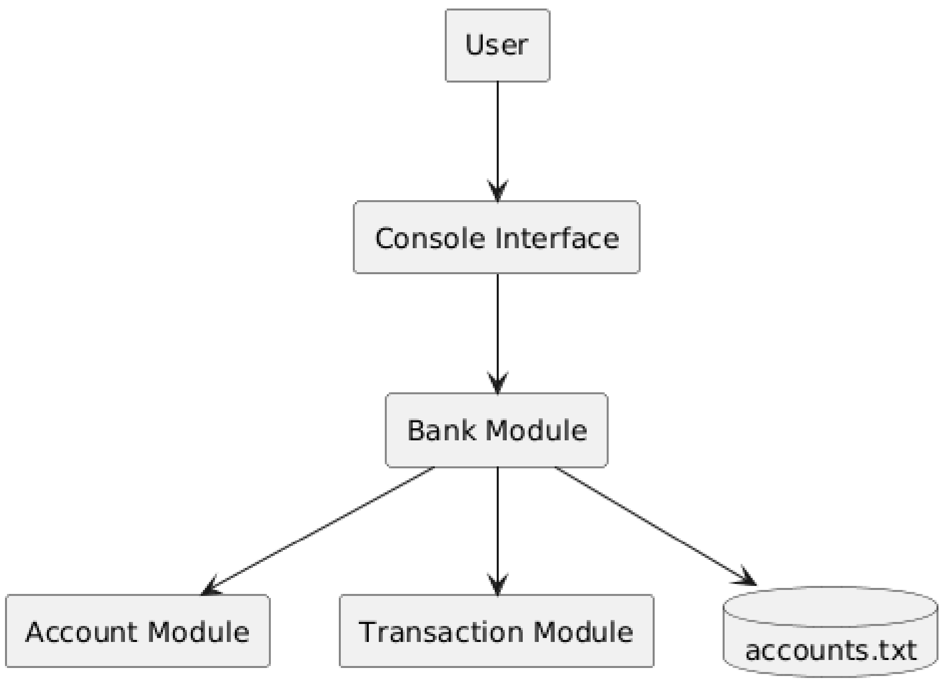


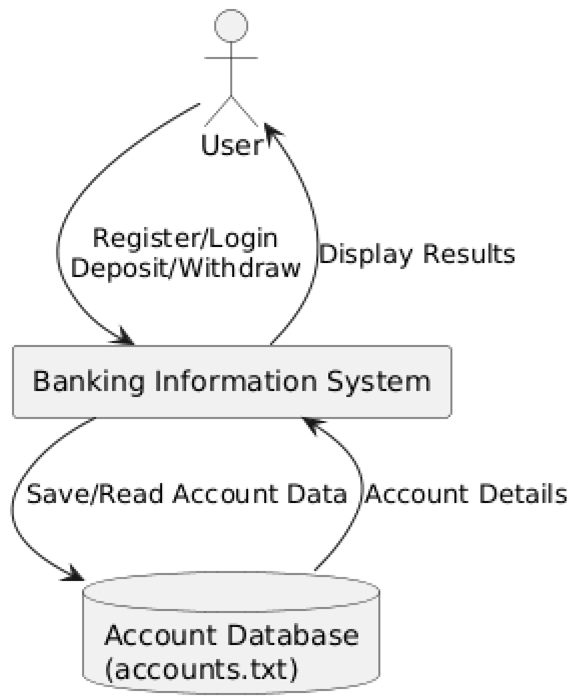
Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces (if applicable)

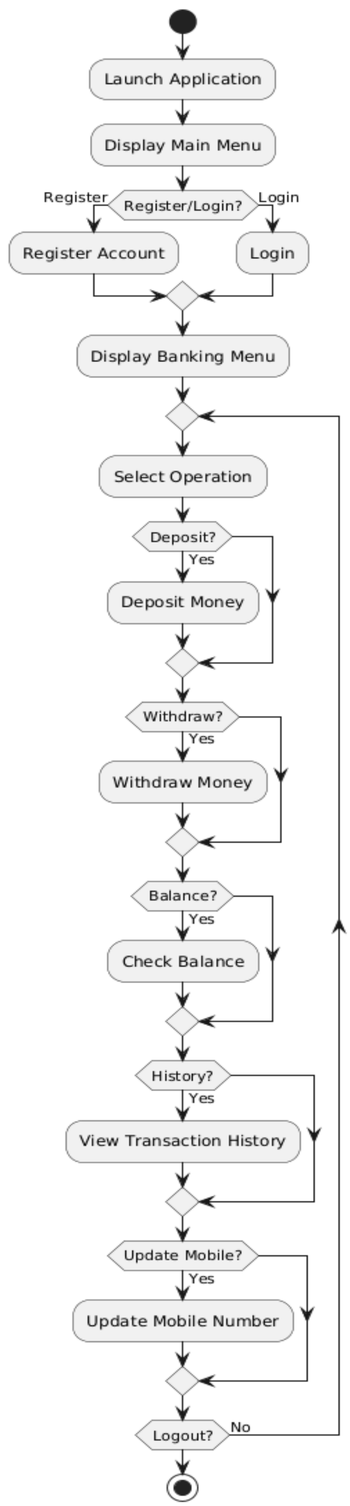
5.3.1 Block Diagram



5.3.2 Data Flow Diagram (Level-0)



5.3.3 Flow Chart



5.3.4 Protocols

Not Applicable. This project is a standalone console-based Core Java application and does not use network communication protocols.

5.3.5 State Machine

Not Applicable. The application is menu-driven and does not implement state machine-based behavior.

5.3.6 Memory Buffer Management

Not Applicable. The project uses Java's built-in memory management and ArrayList for temporary data storage. Manual buffer management is not required.

6 Performance Test.

The Banking Information System was tested by executing all major functionalities under different input conditions to ensure the application performs correctly. Each module was verified using both valid and invalid inputs to check the accuracy of operations and the effectiveness of input validation and exception handling.

The application successfully performed account registration, user login, money deposit, money withdrawal, balance enquiry, transaction history display, and mobile number updates. File handling was also tested to confirm that account details were stored successfully in the text file.

Overall, the application executed all operations correctly without unexpected crashes and produced the expected results for the tested scenarios.

6.1 Test Plan/ Test Cases

Test Case ID	Test Scenario	Expected Result	Status
TC-01	Register a new account	Account created successfully	Pass
TC-02	Login with valid account number	Login successful	Pass
TC-03	Deposit a valid amount	Balance updated successfully	Pass
TC-04	Withdraw a valid amount	Amount withdrawn successfully	Pass
TC-05	Withdraw more than available balance	"Insufficient Balance" message displayed	Pass
TC-06	Check account balance	Current balance displayed correctly	Pass
TC-07	View transaction history	All transactions displayed correctly	Pass
TC-08	Update mobile number	Mobile number updated successfully	Pass
TC-09	Enter an invalid mobile number	Validation message displayed	Pass

Test Case ID	Test Scenario	Expected Result	Status
TC-10	Enter a negative deposit amount	Validation message displayed	Pass

6.2 Test Procedure

1. Launch the Banking Information System application.
2. Register a new account by entering valid customer details.
3. Log in using the generated account number.
4. Perform deposit and withdrawal operations.
5. Verify the updated account balance.
6. View the transaction history.
7. Update the registered mobile number.
8. Test invalid inputs such as incorrect mobile numbers and negative deposit amounts.
9. Exit the application and verify that account details are saved successfully.

6.3 Performance Outcome

The Banking Information System performed successfully during testing. All modules functioned as expected and produced accurate results. Input validation and exception handling prevented invalid operations, while file handling ensured that account information was stored correctly. The application demonstrated stability, reliability, and ease of use throughout the testing process.

7 My learnings

During this internship, I gained valuable practical experience in Core Java programming and software development. Developing the Banking Information System helped me understand how theoretical concepts are applied to real-world applications.

Some of the key learnings from this internship include:

- Improved understanding of Object-Oriented Programming concepts such as classes, objects, encapsulation, and abstraction.
- Gained hands-on experience in implementing file handling for storing account information.
- Learned to use exception handling to improve application reliability and handle invalid user inputs.
- Enhanced problem-solving and debugging skills while developing and testing the project.
- Improved knowledge of Java collections, user input handling, and modular programming.
- Learned to use Git and GitHub for version control and project management.
- Developed confidence in designing, implementing, testing, and documenting a complete Java application.

Overall, this internship strengthened my technical skills and gave me practical exposure to software development practices, which will be beneficial for my future career.

8 Future work scope

The current Banking Information System is a console-based application that performs basic banking operations. In the future, the project can be enhanced by adding more advanced features to improve functionality and user experience.

Possible future enhancements include:

- Developing a graphical user interface (GUI) using Java Swing or JavaFX.
- Integrating a database such as MySQL for secure and efficient data storage.
- Implementing secure user authentication with passwords and encryption.
- Adding fund transfer functionality between accounts.
- Generating account statements and transaction reports.
- Introducing role-based access for customers and administrators.
- Developing a web-based or mobile version of the application.
- Implementing online banking features such as bill payments and account management.

These enhancements would make the Banking Information System more scalable, secure, and suitable for real-world banking applications.